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CAPACITOR, INTEGRATED CIRCUIT, AND MANUFACTURING METHOD OF INTEGRATED CIRCUIT

Abstract

A capacitor includes a first electrode, a second electrode over the first electrode, and a composite storage layer sandwiched between the first electrode and the second electrode. The composite storage layer includes a first storage layer and a second storage layer. The first storage layer is in physical contact with the first electrode. The second storage layer is in physical contact with the second electrode. A conduction band offset of the first storage layer is greater than a conduction band offset of the second storage layer, and a dielectric constant of the second storage layer is greater than a dielectric constant of the first storage layer.

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced a fast-paced growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component or line that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 is a schematic cross-sectional view of an integrated circuit in accordance with some embodiments of the disclosure.

[0004] FIG. 2A to FIG. 2F are cross-sectional views illustrating various stages of a manufacturing method of the capacitor structure in FIG. 1 in accordance with some embodiments of the disclosure.

[0005] FIG. 3A is an enlarged view of a region in FIG. 2C in accordance with some embodiments of the disclosure.

[0006] FIG. 3B is an enlarged view of a region in FIG. 2C in accordance with some alternative embodiments of the disclosure.

[0007] FIG. 4 is a schematic cross-sectional view of an integrated circuit in accordance with some alternative embodiments of the disclosure.

[0008] FIG. 5A to FIG. 5F are cross-sectional views illustrating various stages of a manufacturing method of the capacitor structure in FIG. 4 in accordance with some embodiments of the disclosure.

[0009] FIG. 6A is an enlarged view of a region in FIG. 5D in accordance with some embodiments of the disclosure.

[0010] FIG. 6B is an enlarged view of a region in FIG. 5D in accordance with some alternative embodiments of the disclosure.

[0011] FIG. 7 is a schematic cross-sectional view of an integrated circuit in accordance with some alternative embodiments of the disclosure.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] FIG. 1 is a schematic cross-sectional view of an integrated circuit IC1 in accordance with some embodiments of the disclosure. Referring to FIG. 1, the integrated circuit IC1 includes a substrate 20, an interconnect structure 30, a transistor 40, a passivation layer 50, a post-passivation layer 60, a plurality of conductive pads 70, and a plurality of conductive terminals 80. In some embodiments, the substrate 20 is made of elemental semiconductor materials, such as crystalline silicon, diamond, or germanium; compound semiconductor materials, such as silicon carbide, gallium arsenic, indium arsenide, or indium phosphide; or alloy semiconductor materials, such as silicon germanium, silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. The substrate 20 may be a bulk silicon substrate, a silicon-on-insulator (SOI) substrate, or a germanium-on-insulator (GOI) substrate.

[0015] In some embodiments, the substrate 20 includes various doped regions depending on circuit requirements (e.g., p-type semiconductor substrate or n-type semiconductor substrate). In some embodiments, the doped regions are doped with p-type or n-type dopants. For example, the doped regions may be doped with p-type dopants, such as boron or BF₂; n-type dopants, such as phosphorus or arsenic; and/or combinations thereof. In some embodiments, these doped regions serve as source/drain regions of the transistor 40, which is formed over the substrate 20.

Throughout the disclosure, source/drain region(s) may refer to a source or a drain, individually or collectively dependent upon the context. Depending on the types of the dopants in the doped regions, the transistor 40 may be referred to as n-type transistor or p-type transistor. In some embodiments, the transistor 40 further includes a metal gate and a channel under the metal gate. The channel is located between the source region and the drain region to serve as a path for electrons to travel through when the transistor 40 is turned on. On the other hand, the metal gate is located above the substrate 20 and is embedded in the interconnect structure 30. In some embodiments, the transistor 40 is formed using suitable Front-end-of-line (FEOL) process. That is, the transistor 40 may be referred to as a FEOL transistor. For simplicity, one transistor 40 is shown in FIG. 1. However, it should be understood that more than one transistors 40 may be presented depending on the application of the integrated circuit IC1. When multiple transistors 40 are presented, these transistors 40 may be separated by shallow trench isolation (STI; not shown) located between two adjacent transistors 40. In some embodiments, the FEOL process may be further utilized to form other elements. For example, memory peripheral circuits, Static Random Access Memory (SRAM), and other computing circuits may be formed within/near the substrate 20 through the FEOL processes.

[0016] As illustrated in FIG. 1, the interconnect structure 30 is disposed on the substrate 20. In some embodiments, the interconnect structure 30 includes a plurality of conductive vias 32, a plurality of conductive patterns 34, a plurality of dielectric layers 36, and a capacitor structure C1. As illustrated in FIG. 1, the conductive patterns 34 and the conductive vias 32 are embedded in the dielectric layers 36. In some embodiments, the conductive patterns 34 located at different level heights are connected to one another through the conductive vias 32. In other words, the conductive patterns 34 are electrically connected to one another through the conductive vias 32. In some embodiments, the bottommost conductive vias 32 are connected to the transistor 40. For example, the bottommost conductive vias 32 are connected to the metal gate, which is embedded in the bottommost dielectric layer 36, of the transistor 40. In other words, the bottommost conductive vias 32 establish electrical connection between the transistor 40 and the conductive patterns 34 of the

interconnect structure **30**. It should be noted that in some alternative cross-sectional views, other bottommost conductive vias **32** are also connected to the source/drain regions of the transistor **40**. That is, in some embodiments, the bottommost conductive vias **32** may be referred to as “contact structures” of the transistor **40**.

[0017] In some embodiments, a material of the dielectric layers **36** includes polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzooxazole (PBO), or any other suitable polymer-based dielectric material. Alternatively, the dielectric layers **36** may be formed of oxides or nitrides, such as silicon oxide, silicon nitride, aluminum oxide, hafnium oxide, hafnium zirconium oxide, or the like. In some embodiments, different dielectric layers **36** are formed by the same material. However, the disclosure is not limited thereto. In some alternative embodiments, different dielectric layers **36** may be formed by different materials. The dielectric layers **36** may be formed by suitable fabrication techniques, such as spin-on coating, chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), or the like.

[0018] In some embodiments, a material of the conductive patterns **34** and the conductive vias **32** includes aluminum, titanium, copper, nickel, tungsten, or alloys thereof. The conductive patterns **34** and the conductive vias **32** may be formed by electroplating, deposition, and/or photolithography and etching. In some embodiments, the conductive patterns **34** and the underlying conductive vias **32** are formed simultaneously. It should be noted that the number of the dielectric layers **36**, the number of the conductive patterns **34**, and the number of the conductive vias **32** illustrated in FIG. **1** are merely for illustrative purposes, and the disclosure is not limited thereto. In some alternative embodiments, fewer or more layers of the dielectric layers **36**, the conductive patterns **34**, and/or the conductive vias **32** may be formed depending on the circuit design.

[0019] As illustrated in FIG. **1**, the passivation layer **50**, the conductive pads **70**, the post-passivation layer **60**, and the conductive terminals **80** are sequentially formed on the interconnect structure **30**. In some embodiments, the passivation layer **50** is disposed on the topmost dielectric layer **36** and the topmost conductive patterns **34**. In some embodiments, the passivation layer **50** has a plurality of openings partially exposing each topmost conductive pattern **34**. In some embodiments, the passivation layer **50** is a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, or a dielectric layer formed by other suitable dielectric materials. The passivation layer **50** may be formed by suitable fabrication techniques, such as high-density-plasma chemical vapor deposition (HDP-CVD), PECVD, or the like.

[0020] In some embodiments, the conductive pads **70** are formed over the passivation layer **50**. In some embodiments, the conductive pads **70** extend into the openings of the passivation layer **50** to be in physical contact with the topmost conductive patterns **34**. That is, the conductive pads **70** are electrically connected to the interconnect structure **30**. In some embodiments, the conductive pads **70** include aluminum pads, copper pads, titanium pads, nickel pads, tungsten pads, or other suitable metal pads. The conductive pads **70** may be formed by, for example, electroplating, deposition, and/or photolithography and etching. It should be noted that the number and the shape of the conductive pads **70** illustrated in FIG. **1** are merely for illustrative purposes, and the disclosure is not limited thereto. In some alternative embodiments, the number and the shape of the conductive pad **70** may be adjusted based on demand.

[0021] In some embodiments, the post-passivation layer **60** is formed over the passivation layer **50** and the conductive pads **70**. In some embodiments, the post-passivation layer **60** is formed on the conductive pads **70** to protect the conductive pads **70**. In some embodiments, the post-passivation layer **60** has a plurality of contact openings partially exposing each conductive pad **70**. The post-passivation layer **60** may be a polyimide layer, a PBO layer, or a dielectric layer formed by other suitable polymers. In some embodiments, the post-passivation layer **60** is formed by suitable fabrication techniques, such as HDP-CVD, PECVD, or the like.

[0022] As illustrated in FIG. **1**, the conductive terminals **80** are formed over the post-passivation layer **60** and the conductive pads **70**. In some embodiments, the conductive terminals **80** extend

into the contact openings of the post-passivation layer **60** to be in physical contact with the corresponding conductive pad **70**. That is, the conductive terminals **80** are electrically connected to the interconnect structure **30** through the conductive pads **70**. In some embodiments, the conductive terminals **80** are conductive pillars, conductive posts, conductive balls, conductive bumps, or the like. In some embodiments, a material of the conductive terminals **80** includes a variety of metals, metal alloys, or metals and mixture of other materials. For example, the conductive terminals **80** may be made of aluminum, titanium, copper, nickel, tungsten, tin, and/or alloys thereof. The conductive terminals **80** are formed by, for example, deposition, electroplating, screen printing, or other suitable methods. In some embodiments, the conductive terminals **80** are used to establish electrical connection with other components (not shown) subsequently formed or provided.

[0023] As illustrated in FIG. **1**, the capacitor structure **C1** is embedded in the interconnect structure **30**. For example, the capacitor structure **C1** is embedded in one of the dielectric layers **36**. In some embodiments, the capacitor structure **C1** is electrically connected to the underlying and overlying conductive vias **32**/conductive patterns **34**. The formation method and the structure of the capacitor structure **C1** will be described below in conjunction with FIG. **2A** to FIG. **2F**.

[0024] FIG. **2A** to FIG. **2F** are cross-sectional views illustrating various stages of a manufacturing method of the capacitor structure **C1** in FIG. **1** in accordance with some embodiments of the disclosure. Referring to FIG. **2A**, a sub-dielectric layer **110** is provided. In some embodiments, the sub-dielectric layer **110** is part of one of the dielectric layers **36** of the interconnect structure **30** in FIG. **1**, so the detailed description thereof is omitted herein. In some embodiments, a plurality of conductive patterns **200** is formed in the sub-dielectric layer **110**. For example, the conductive patterns **200** are embedded in the sub-dielectric layer **110**. In some embodiments, the conductive patterns **200** are electrically connected to the conductive patterns **34** and the conductive vias **32** of the interconnect structure **30** in FIG. **1**. In some embodiments, each of the conductive patterns **200** includes a seed layer and a metal layer on the seed layer. The seed layer may be a conformal layer lining sidewalls and a bottom surface of the metal layer. In some embodiments, the seed layer may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. For example, the seed layer is a titanium/copper composited layer. The metal layer may include a metal material, such as Al, Cu, W, Co, Pd, Pt, Ni, other low resistivity metal constituent, an alloy thereof, or a combination thereof, and are formed by an electroplating process. As illustrated in FIG. **2A**, a top surface of the sub-dielectric layer **110** and top surfaces of the conductive patterns **200** are coplanar.

[0025] Referring to FIG. **2B**, a sub-dielectric layer **120** is formed on the sub-dielectric layer **110** and the conductive patterns **200**. In some embodiments, the sub-dielectric layer **120** is part of one of the dielectric layers **36** of the interconnect structure **30** in FIG. **1**. In some embodiments, a material and a formation method of the sub-dielectric layer **120** are similar to that of the sub-dielectric layer **110**, so the detailed descriptions thereof are omitted herein.

[0026] Referring to FIG. **2C**, a capacitor **300** is formed on the sub-dielectric layer **120**. In some embodiments, the capacitor **300** includes an electrode **310**, a composite storage layer **320**, an electrode **330**, a composite storage layer **340**, and an electrode **350**. In some embodiments, the capacitor **300** is formed by the following steps. First, a first electrode material layer (not shown) is conformally formed on a top surface of the sub-dielectric layer **120** by a suitable technique such as a physical vapor deposition (PVD) process or a plating process. Thereafter, the first electrode material layer is patterned by a photolithography and etching process to form the electrode **310**. As illustrated in FIG. **2C**, the electrode **310** covers a first portion of the sub-dielectric layer **120** and exposes a second portion of the sub-dielectric layer **120**. After the electrode **310** is formed on the sub-dielectric layer **120**, the composite storage layer **320** is conformally formed on the electrode **310** and the sub-dielectric layer **120**. For example, the composite storage layer **320** covers the electrode **310** and the second portion of the sub-dielectric layer **120** that is exposed by the electrode **310**. In some embodiments, the composite storage layer **320** is formed by, for example, CVD,

PECVD, flowable chemical vapor deposition (FCVD), HDP-CVD, sub-atmospheric chemical vapor deposition (SACVD), PVD, or atomic layer deposition (ALD). Then, a second electrode material layer (not shown) is conformally formed on a top surface of the composite storage layer **320** by a suitable technique such as a PVD process or a plating process. Thereafter, the second electrode material layer is patterned by a photolithography and etching process to form the electrode **330**. As illustrated in FIG. 2C, the electrode **330** covers a first portion of the composite storage layer **320** and exposes a second portion of the composite storage layer **320**. After the electrode **330** is formed on the composite storage layer **320**, the composite storage layer **340** is formed on the electrode **330**. As illustrated in FIG. 2C, the composite storage layer **340** also exposes the second portion of the composite storage layer **320**. In some embodiments, the composite storage layer **340** is formed by, for example, CVD, PECVD, FCVD, HDP-CVD, SACVD, PVD, or ALD. Subsequently, a third electrode material layer (not shown) is conformally formed on a top surface of the second portion of the composite storage layer **320** and a top surface of the composite storage layer **340** by a suitable technique such as a PVD process or a plating process. Lastly, the third electrode material layer is patterned by a photolithography and etching process to form the electrode **350**. As illustrated in FIG. 2C, the electrode **350** covers the second portion of the composite storage layer **320** and a first portion of the composite storage layer **340**. Meanwhile, the electrode **350** exposes a second portion of the composite storage layer **340**. [0027] As illustrated in FIG. 2C, the second portion of the composite storage layer **320** is in physical contact with both the electrode **310** and the electrode **350** to form a metal-insulator-metal (MIM) structure. Meanwhile, the first portion of the composite storage layer **320** is in physical contact with both the electrode **310** and the electrode **330** to form another MIM structure. Moreover, the composite storage layer **340** is in physical contact with both the electrode **330** and the electrode **350** to form yet another MIM structure. That is, the capacitor **300** may be referred to as a MIM capacitor. In some embodiments, the composite storage layers **320** and **340** are utilized to store capacitance.

[0028] In some embodiments, the composite storage layer **320** and the composite storage layer **340** are respectively a multi-layered structure. The detail configuration of the electrode **310**, the composite storage layer **320**, the electrode **330**, the composite storage layer **340**, and the electrode **350** will be discussed below in conjunction with FIG. 3A and FIG. 3B.

[0029] FIG. 3A is an enlarged view of a region R1 in FIG. 2C in accordance with some embodiments of the disclosure. Referring to FIG. 3A, the composite storage layer **320** includes a storage layer **322** and a storage layer **324**. Meanwhile, the composite storage layer **340** includes a storage layer **342** and a storage layer **344**. As illustrated in FIG. 3A, the electrode **310**, the storage layer **322**, the storage layer **324**, the electrode **330**, the storage layer **344**, the storage layer **342**, and the electrode **350** are stacked in sequential order. That is, the composite storage layer **320** is sandwiched between the electrode **310** and the electrode **330**, and the composite storage layer **340** is sandwiched between the electrode **330** and the electrode **350**.

[0030] In some embodiments, the electrodes **310**, **330**, and **350** may respectively be a single-layered structure or a multi-layered structure. In some embodiments, the electrodes **310**, **330**, and **350** respectively include various conductive materials, such as a metal, a metal alloy, a metal nitride, a metal silicide, a metal oxide, graphene, or a combination thereof. For example, the electrodes **310**, **330**, and **350** respectively include aluminum (Al), titanium (Ti), copper (Cu), tungsten (W), platinum (Pt), palladium (Pd), osmium (Os), ruthenium (Ru), tantalum (Ta), or an alloy thereof, titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), molybdenum nitride (MON), TaSiN, TiSiN, WSiN, tungsten silicide, titanium silicide, cobalt silicide, zirconium silicide, platinum silicide, molybdenum silicide, copper silicide, nickel silicide, indium tin oxide (ITO), iridium oxide (IrO₂), rhenium oxide (ReO₂), rhenium trioxide (ReO₃), or a combination thereof. In some embodiments, the material of the electrodes **310**, **330**, and **350** may be the same as one another or may be different from one another.

[0031] As illustrated in FIG. 3A, the storage layer 322 is in physical contact with the electrode 310. Meanwhile, the storage layer 324 is in physical contact with the electrode 330. Moreover, the storage layer 322 is in physical contact with the storage layer 324. That is, the storage layer 324 is located between the storage layer 322 and the electrode 330. In some embodiments, a conduction band offset (conduction barrier) of the storage layer 322 is different from a conduction band offset of the storage layer 324. Moreover, a dielectric constant (k-value) of the storage layer 322 is also different from a dielectric constant of the storage layer 324. For example, the conduction band offset of the storage layer 322 is greater than the conduction band offset of the storage layer 324, and the dielectric constant of the storage layer 324 is greater than the dielectric constant of the storage layer 322. In order to satisfy the foregoing constraint, a material of the storage layer 322 is different from a material of the storage layer 324. For example, the material of the storage layer 322 includes LaO, AlO, SiO, HfO having a dopant, ZrO having a dopant, or a combination thereof. Meanwhile, the material of the storage layer 324 includes ZrO, HfO, HfZrO, HfO having a dopant, ZrO having a dopant, or a combination thereof. The dopant in the storage layer 322 and the dopant in the storage layer 324 include La, Al, Si, or a combination. In some embodiments, the dopant in the storage layer 322 and the dopant the storage layer 324 are the same. Nevertheless, a doping concentration of the dopant in the storage layer 322 is different form a doping concentration of the dopant in the storage layer 324. For example, the doping concentration of the dopant in the storage layer 322 is greater than the doping concentration of the dopant in the storage 324. However, the disclosure is not limited thereto. In some alternative embodiments, the dopant in the storage layer 322 is different from the dopant in the storage layer 324. It should be noted that the material of the storage layer 322 and the material of the storage layer 324 are not particular limited, as long as the conduction band offset of the storage layer 322 is greater than the conduction band offset of the storage layer 324 and the dielectric constant of the storage layer 324 is greater than the dielectric constant of the storage layer 322.

[0032] In some embodiments, the storage layer 342 is in physical contact with the electrode 350. Meanwhile, the storage layer 344 is in physical contact with the electrode 330. Moreover, the storage layer 342 is in physical contact with the storage layer 344. That is, the storage layer 344 is located between the storage layer 342 and the electrode 330. In some embodiments, a conduction band offset of the storage layer 342 is different from a conduction band offset of the storage layer 344. Moreover, a dielectric constant of the storage layer 342 is also different from a dielectric constant of the storage layer 344. For example, the conduction band offset of the storage layer 342 is greater than the conduction band offset of the storage layer 344, and the dielectric constant of the storage layer 344 is greater than the dielectric constant of the storage layer 342. In order to satisfy the foregoing constraint, a material of the storage layer 342 is different from a material of the storage layer 344. In some embodiments, the material of the storage layer 342 and the material of the storage layer 344 are respectively similar to that of the storage layer 322 and the storage layer 324, so the detailed descriptions thereof are omitted herein. It should be noted that the material of the storage layer 342 and the material of the storage layer 344 are not particular limited, as long as the conduction band offset of the storage layer 342 is greater than the conduction band offset of the storage layer 344 and the dielectric constant of the storage layer 344 is greater than the dielectric constant of the storage layer 342.

[0033] In some embodiments, during operation of the capacitor 300, unipolar operation voltage is applied to the electrodes 310, 330, and 350. For example, a first voltage is applied to the electrode 310, a second voltage is applied to the electrode 330, and a third voltage is applied to the electrode 350. In some embodiments, the second voltage is greater than the first voltage. Meanwhile, the second voltage is also greater than the third voltage. For example, the first voltage is a negative voltage, the second voltage is a positive voltage, and the third voltage is a negative voltage. That is, the electrode 310 and the electrode 350 are electrically connected to a negative voltage while the electrode 330 is electrically connected to a positive voltage. In some embodiments, since the

conduction band offset of the storage layers (i.e., the storage layer 322 and the storage layer 342) contacting the electrodes (i.e., the electrode 310 and the electrode 350) applied with negative voltage is greater than the conduction band offset of the storage layers (i.e., the storage layer 324 and the storage layer 344) contacting the electrode (i.e., the electrode 330) applied with positive voltage, phenomenon of electron tunneling may be sufficiently alleviated, and the leakage of electric current in the capacitor 300 may be sufficiently reduced. Moreover, since the dielectric constant of the storage layers (i.e., the storage layer 324 and the storage layer 344) contacting the electrode (i.e., the electrode 330) applied with positive voltage is greater than the dielectric constant of the storage layers (i.e., the storage layer 322 and the storage layer 342) contacting the electrodes (i.e., the electrode 310 and the electrode 350) applied with negative voltage, the overall capacitance of the capacitor 300 may be sufficiently increased. Furthermore, since the material of the storage layer 324 and the storage layer 344 has lower H.sub.2O reactivity, degradation of k-value caused by reaction with H.sub.2O in the ambient environment may be sufficiently suppressed, thereby ensuring the reliability of the capacitor 300.

[0034] It should be noted that the detail configuration of the electrode 310, the composite storage layer 320, the electrode 330, the composite storage layer 340, and the electrode 350 in FIG. 3A is merely an exemplary illustration, and the disclosure is not limited thereto. These layers may be arranged in other configurations as shown in FIG. 3B.

[0035] FIG. 3B is an enlarged view of a region R1 in FIG. 2C in accordance with some alternative embodiments of the disclosure. Referring to FIG. 3B, the composite storage layer 320 includes a storage layer 322, a storage layer 324, and a blocking layer 326. Meanwhile, the composite storage layer 340 includes a storage layer 342, a storage layer 344, and a blocking layer 346. In some embodiments, the blocking layer 326 is sandwiched between the storage layer 322 and the storage layer 324, and the blocking layer 346 is sandwiched between the storage layer 342 and the storage layer 344. That is, the blocking layer 326 is deposited on the storage layer 322 prior to the deposition of the storage layer 324, and the blocking layer 346 is deposited on the storage layer 344 prior to the deposition of the storage layer 342. As illustrated in FIG. 3B, the electrode 310, the storage layer 322, the blocking layer 326, the storage layer 324, the electrode 330, the storage layer 344, the blocking layer 346, the storage layer 342, and the electrode 350 are stacked in sequential order. That is, the composite storage layer 320 is sandwiched between the electrode 310 and the electrode 330, and the composite storage layer 340 is sandwiched between the electrode 330 and the electrode 350.

[0036] In some embodiments, the electrode 310, the storage layer 322, the storage layer 324, the electrode 330, the storage layer 342, the storage layer 344, and the electrode 350 in FIG. 3B are respectively similar to the electrode 310, the storage layer 322, the storage layer 324, the electrode 330, the storage layer 342, the storage layer 344, and the electrode 350 in FIG. 3A, so the detailed descriptions thereof are omitted herein.

[0037] In some embodiments, a dielectric constant of the blocking layer 326 is greater than the dielectric constant of the storage layer 322 and the dielectric constant of the storage layer 324. For example, the dielectric constant of the blocking layer 326 is greater than 10. In some embodiments, a crystallinity of the blocking layer 326 is also greater than a crystallinity of the storage layer 322 and a crystallinity of the storage layer 324. In order to satisfy the foregoing constraint, a material of the blocking layer 326 is different from the material of the storage layer 322 and the material of the storage layer 324. For example, the material of the blocking layer 326 includes AlO, Y.sub.2O.sub.3, HfO having a dopant, ZrO having a dopant, or a combination thereof. The dopant in the blocking layer 326 includes La, Al, Si, or a combination. In some embodiments, the dopant in the blocking layer 326 is the same as the dopant in the storage layer 322 and the dopant in the storage layer 324. Nevertheless, a doping concentration of the dopant in the blocking layer 326 is greater than the doping concentration of the dopant in the storage layer 322 and the doping concentration of the dopant in the storage layer 324. However, the disclosure is not limited thereto.

In some alternative embodiments, the dopant in the blocking layer **326** is different from the dopant in the storage layer **322** and the dopant in the storage layer **324**. It should be noted that the material of the blocking layer **326** is not particular limited, as long as the dielectric constant and the crystallinity of the blocking layer **326** are greater than the dielectric constant and the crystallinity of the storage layer **322** and the dielectric constant and the crystallinity of the storage layer **324**.

[0038] In some embodiments, a material of the blocking layer **346** is similar to that of the blocking layer **326**, so the detailed description thereof is omitted herein. It should be noted that the material of the blocking layer **346** is not particular limited, as long as the dielectric constant and the crystallinity of the blocking layer **346** are greater than the dielectric constant and the crystallinity of the storage layer **342** and the dielectric constant and the crystallinity of the storage layer **344**.

[0039] In some embodiments, since the blocking layer **326** is sandwiched between the storage layer **322** and the storage layer **324**, the blocking layer **326** is able to serve as a crystalline interruption layer for the storage layers **322** and **324**. Similarly, since the blocking layer **346** is sandwiched between the storage layer **342** and the storage layer **344**, the blocking layer **346** is able to serve as a crystalline interruption layer for the storage layers **342** and **344**. That is, the blocking layers **326** and **346** may effectively suppress the storage layers **322**, **324**, **342**, and **344** from crystallization, thereby increasing the overall capacitance of the capacitor **300**.

[0040] Referring back to FIG. 2D, after the capacitor **300** is formed, a sub-dielectric layer **130** is formed on the capacitor **300**. In some embodiments, the sub-dielectric layer **130** is part of one of the dielectric layers **36** of the interconnect structure **30** in FIG. 1. In some embodiments, a material and a formation method of the sub-dielectric layer **130** are similar to that of the sub-dielectric layer **110**, so the detailed descriptions thereof are omitted herein. As illustrated in FIG. 2D, the sub-dielectric layer **130** is in physical contact with the electrode **350** and the composite storage layer **340**.

[0041] Referring to FIG. 2E, a portion of the sub-dielectric layer **120**, a portion of the capacitor **300**, and a portion of the sub-dielectric layer **130** are removed to form a plurality of openings OP1. For example, the sub-dielectric layer **120**, the electrode **310**, the composite storage layer **320**, the electrode **330**, the composite storage layer **340**, the electrode **350**, and the sub-dielectric layer **130** are patterned by a patterning process to form the openings OP1. The patterning process includes photolithography and one or more etching processes, for example. In some embodiments, the openings OP1 penetrate through the sub-dielectric layer **130**, the capacitor **300**, and the sub-dielectric layer **120** to expose the underlying conductive patterns **200**.

[0042] Referring to FIG. 2F, a plurality of conductive features **400** is formed in the openings OP1 and on the sub-dielectric layer **130** to obtain the capacitor structure C1. In some embodiments, each conductive feature **400** includes a conductive via **402** and a conductive line **404** disposed on the conductive via **402**. As illustrated in FIG. 2F, the conductive vias **402** fill up the openings OP1. Meanwhile, the conductive lines **404** are disposed on top of the sub-dielectric layer **130**. In some embodiments, each of the conductive features **400** includes a seed layer and a metal layer on the seed layer. The seed layer may be a conformal layer lining sidewalls and a bottom surface of the metal layer. In some embodiments, the seed layer may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. For example, the seed layer is a titanium/copper composited layer. The metal layer may include a metal material, such as Al, Cu, W, Co, Pd, Pt, Ni, other low resistivity metal constituent, an alloy thereof, or a combination thereof, and are formed by an electroplating process. In some embodiments, the conductive features **400** are electrically connected to the conductive patterns **34** and the conductive vias **32** of the interconnect structure **30** in FIG. 1.

[0043] As illustrated in FIG. 2F, the capacitor **300** is located aside the conductive features **400**. For example, the capacitor **300** is located aside the conductive vias **402** of the conductive features **400**. In some embodiments, the electrode **310** and the electrode **350** are in physical contact with sidewalls of one of the conductive vias **402**. On the other hand, the electrode **330** is in physical

contact with sidewalls of one of the conductive vias **402**. Meanwhile, the electrode **310** and the electrode **350** are electrically isolated from the electrode **330**. In some embodiments, the capacitor **300** is electrically connected to the conductive patterns **200** through the conductive features **400**. That is, the capacitor **300** is connected to the conductive patterns **34** and the conductive vias **32** of the interconnect structure **30** in FIG. **1** through the conductive features **400** and/or the conductive patterns **200**.

[0044] In some embodiments, since the interconnect structure **30** is being considered as formed during back-end-of-line (BEOL) process, the capacitor structure **C1** is also being considered as formed during BEOL processes.

[0045] FIG. **4** is a schematic cross-sectional view of an integrated circuit **IC2** in accordance with some alternative embodiments of the disclosure. Referring to FIG. **4**, the integrated circuit **IC2** in FIG. **4** is similar to the integrated circuit **IC1** in FIG. **1**, so the detailed description thereof is omitted herein. The difference between the integrated circuit **IC2** in FIG. **4** and the integrated circuit **IC1** in FIG. **1** lies in that the capacitor structure **C1** in FIG. **1** is replaced by a capacitor structure **C2** in FIG. **4**.

[0046] As illustrated in FIG. **4**, the capacitor structure **C2** is embedded in the interconnect structure **30**. For example, the capacitor structure **C2** is embedded in one of the dielectric layers **36**. In some embodiments, the capacitor structure **C2** is electrically connected to the underlying and overlaying conductive vias **32**/conductive patterns **34**. The formation method and the structure of the capacitor structure **C2** will be described below in conjunction with FIG. **5A** to FIG. **5F**.

[0047] FIG. **5A** to FIG. **5F** are cross-sectional views illustrating various stages of a manufacturing method of the capacitor structure **C2** in FIG. **4** in accordance with some embodiments of the disclosure. Referring to FIG. **5A**, a sub-dielectric layer **510** is provided. In some embodiments, the sub-dielectric layer **510** is part of one of the dielectric layers **36** of the interconnect structure **30** in FIG. **1**, so the detailed description thereof is omitted herein. In some embodiments, a conductive pattern **600** is formed in the sub-dielectric layer **510**. For example, the conductive pattern **600** is embedded in the sub-dielectric layer **510**. In some embodiments, the conductive pattern **600** is electrically connected to the conductive patterns **34** and the conductive vias **32** of the interconnect structure **30** in FIG. **1**. In some embodiments, the conductive pattern **600** includes a seed layer and a metal layer on the seed layer. The seed layer may be a conformal layer lining sidewalls and a bottom surface of the metal layer. In some embodiments, the seed layer may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. For example, the seed layer is a titanium/copper composited layer. The metal layer may include a metal material, such as Al, Cu, W, Co, Pd, Pt, Ni, other low resistivity metal constituent, an alloy thereof, or a combination thereof, and are formed by an electroplating process. As illustrated in FIG. **5A**, a top surface of the sub-dielectric layer **510** and a top surface of the conductive pattern **600** are coplanar.

[0048] Referring to FIG. **5B**, an etch stop layer **700** and a sub-dielectric layer **520** are sequentially formed on the sub-dielectric layer **510** and the conductive pattern **600**. In some embodiments, the etch stop layer **700** is in physical contact with the sub-dielectric layer **510** and the conductive pattern **600**. For example, the etch stop layer **700** is sandwiched between the sub-dielectric layer **510** and the sub-dielectric layer **520** and is sandwiched between the conductive pattern **600** and the sub-dielectric layer **520**. In some embodiments, a material of the etch stop layer **700** includes silicon carbide, silicon nitride, or the like. In some embodiments, the etch stop layer **700** is formed by any suitable method, such as CVD, spin-on coating, or the like. In some embodiments, the sub-dielectric layer **520** is part of one of the dielectric layers **36** of the interconnect structure **30** in FIG. **1**. In some embodiments, a material and a formation method of the sub-dielectric layer **520** are similar to that of the sub-dielectric layer **510**, so the detailed descriptions thereof are omitted herein.

[0049] Referring to FIG. **5C**, a portion of the sub-dielectric layer **520** and a portion of the etch stop layer **700** are removed to form a plurality of openings **OP2**. For example, the sub-dielectric layer

520 and the etch stop layer **700** are patterned by a patterning process to form the openings **OP2**. The patterning process includes photolithography and one or more etching processes, for example. In some embodiments, the openings **OP2** penetrate through the sub-dielectric layer **520** and the etch stop layer **700** to expose the underlying conductive pattern **600**.

[0050] Referring to FIG. 5D, a capacitor **800** is formed in the openings **OP2** and on the sub-dielectric layer **520**. In some embodiments, the capacitor **800** includes an electrode **810**, a composite storage layer **820**, and an electrode **830**. In some embodiments, the capacitor **800** is formed by the following steps. First, a first electrode material layer (not shown) is conformally formed on the sub-dielectric layer **520**, the etch stop layer **700**, and the conductive pattern **600** by a suitable technique such as a PVD process or a plating process. That is, a portion of the first electrode material layer is disposed on a top surface of the sub-dielectric layer **520** while another portion of the first electrode material layer extends into the openings **OP2** to be in physical contact with the conductive pattern **600**. Thereafter, a composite storage material layer (not shown) is conformally formed on the first electrode material layer. In some embodiments, the composite storage material layer is formed by, for example, CVD, PECVD, FCVD, HDP-CVD, SACVD, PVD, or ALD. In some embodiments, a portion of the composite storage material layer is disposed over the sub-dielectric layer **520** while another portion of the composite storage material layer extends into the openings **OP2**. Subsequently, a second electrode material layer (not shown) is conformally formed on the composite storage material layer by a suitable technique such as a PVD process or a plating process. In some embodiments, a portion of the second electrode material layer is disposed over the sub-dielectric layer **520** while another portion of the second electrode material layer extends into the openings **OP2**. Lastly, the first electrode material layer, the composite storage material layer, and the second electrode material layer are patterned by a photolithography and etching process to respectively form the electrode **810**, the composite storage layer **820**, and the electrode **830**. In some embodiments, the electrode **810** and the composite storage layer **820** are patterned by the same photomask. Meanwhile, the electrode **830** is patterned by another photomask. As such, sidewalls of tails of the electrode **810** are aligned with sidewalls of tails of the composite storage layer **820**, as shown in FIG. 5D. Meanwhile, sidewalls of tails of the electrode **830** are not aligned with the sidewalls of the tails of the electrode **810** and the sidewalls of the tails of the composite storage layer **820**. For example, the electrode **830** exposes a portion of a top surface of the composite storage layer **820**.

[0051] In some embodiments, the capacitor **800** is partially located in the openings **OP2**. Therefore, the capacitor **800** may be referred to as a deep trench capacitor (DTC). As illustrated in FIG. 5D, the composite storage layer **820** is in physical contact with both the electrode **810** and the electrode **830** to form a MIM structure. That is, the capacitor **800** may be referred to as a MIM capacitor. In some embodiments, the composite storage layer **820** is utilized to store capacitance.

[0052] In some embodiments, the composite storage layer **820** is a multi-layered structure. The detail configuration of the electrode **810**, the composite storage layer **820**, and the electrode **830** will be discussed below in conjunction with FIG. 6A and FIG. 6B.

[0053] FIG. 6A is an enlarged view of a region **R2** in FIG. 5D in accordance with some embodiments of the disclosure. Referring to FIG. 6A, the composite storage layer **820** includes a storage layer **822** and a storage layer **824**. As illustrated in FIG. 6A, the electrode **810**, the storage layer **822**, the storage layer **824**, and the electrode **830** are stacked in sequential order. That is, the composite storage layer **820** is sandwiched between the electrode **810** and the electrode **830**.

[0054] In some embodiments, the electrodes **810** and **830** may respectively be a single-layered structure or a multi-layered structure. In some embodiments, the electrodes **810** and **830** respectively include various conductive materials, such as a metal, a metal alloy, a metal nitride, a metal silicide, a metal oxide, graphene, or a combination thereof. For example, the electrodes **810** and **830** respectively include aluminum (Al), titanium (Ti), copper (Cu), tungsten (W), platinum (Pt), palladium (Pd), osmium (Os), ruthenium (Ru), tantalum (Ta), or an alloy thereof, titanium

nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), molybdenum nitride (MON), TaSiN, TiSiN, WSiN, tungsten silicide, titanium silicide, cobalt silicide, zirconium silicide, platinum silicide, molybdenum silicide, copper silicide, nickel silicide, indium tin oxide (ITO), iridium oxide (IrO.sub.2), rhenium oxide (ReO.sub.2), rhenium trioxide (ReO.sub.3), or a combination thereof. In some embodiments, the material of the electrodes **810** and **830** may be the same as each other or may be different from each other.

[0055] As illustrated in FIG. **6A**, the storage layer **822** is in physical contact with the electrode **810**. Meanwhile, the storage layer **824** is in physical contact with the electrode **830**. Moreover, the storage layer **822** is in physical contact with the storage layer **824**. That is, the storage layer **824** is located between the storage layer **822** and the electrode **830**. In some embodiments, a conduction band offset of the storage layer **822** is different from a conduction band offset of the storage layer **824**. Moreover, a dielectric constant of the storage layer **822** is also different from a dielectric constant of the storage layer **824**. For example, the conduction band offset of the storage layer **822** is greater than the conduction band offset of the storage layer **824**, and the dielectric constant of the storage layer **824** is greater than the dielectric constant of the storage layer **822**. In order to satisfy the foregoing constraint, a material of the storage layer **822** is different from a material of the storage layer **824**. For example, the material of the storage layer **822** includes LaO, AlO, SiO, HfO having a dopant, ZrO having a dopant, or a combination thereof. Meanwhile, the material of the storage layer **824** includes ZrO, HfO, HfZrO, HfO having a dopant, ZrO having a dopant, or a combination thereof. The dopant in the storage layer **822** and the dopant in the storage layer **824** include La, Al, Si, or a combination. In some embodiments, the dopant in the storage layer **822** and the dopant in the storage layer **824** are the same. Nevertheless, a doping concentration of the dopant in the storage layer **822** is different from a doping concentration of the dopant in the storage layer **824**. For example, the doping concentration of the dopant in the storage layer **822** is greater than the doping concentration of the dopant in the storage layer **824**. However, the disclosure is not limited thereto. In some alternative embodiments, the dopant in the storage layer **822** is different from the dopant in the storage layer **824**. It should be noted that the material of the storage layer **822** and the material of the storage layer **824** are not particularly limited, as long as the conduction band offset of the storage layer **822** is greater than the conduction band offset of the storage layer **824** and the dielectric constant of the storage layer **824** is greater than the dielectric constant of the storage layer **822**.

[0056] In some embodiments, during operation of the capacitor **800**, unipolar operation voltage is applied to the electrodes **810** and **830**. For example, a first voltage is applied to the electrode **810** and a second voltage is applied to the electrode **830**. In some embodiments, the second voltage is greater than the first voltage. For example, the first voltage is a negative voltage, and the second voltage is a positive voltage. That is, the electrode **810** is electrically connected to a negative voltage while the electrode **830** is electrically connected to a positive voltage. In some embodiments, since the conduction band offset of the storage layer (i.e., the storage layer **822**) contacting the electrode (i.e., the electrode **810**) applied with negative voltage is greater than the conduction band offset of the storage layer (i.e., the storage layer **824**) contacting the electrode (i.e., the electrode **830**) applied with positive voltage, phenomenon of electron tunneling may be sufficiently alleviated, and the leakage of electric current in the capacitor **800** may be sufficiently reduced. Moreover, since the dielectric constant of the storage layer (i.e., the storage layer **824**) contacting the electrode (i.e., the electrode **830**) applied with positive voltage is greater than the dielectric constant of the storage layer (i.e., the storage layer **822**) contacting the electrode (i.e., the electrode **810**) applied with negative voltage, the overall capacitance of the capacitor **800** may be sufficiently increased. Furthermore, since the material of the storage layer **824** has lower H.sub.2O reactivity, degradation of k-value caused by reaction with H.sub.2O in the ambient environment may be sufficiently suppressed, thereby ensuring the reliability of the capacitor **800**.

[0057] It should be noted that the detail configuration of the electrode **810**, the composite storage

layer **820**, and the electrode **830** in FIG. 6A is merely an exemplary illustration, and the disclosure is not limited thereto. These layers may be arranged in other configurations as shown in FIG. 6B. [0058] FIG. 6B is an enlarged view of a region R2 in FIG. 5D in accordance with some alternative embodiments of the disclosure. Referring to FIG. 6B, the composite storage layer **820** includes a storage layer **822**, a storage layer **824**, and a blocking layer **826**. In some embodiments, the blocking layer **826** is sandwiched between the storage layer **822** and the storage layer **824**. That is, the blocking layer **826** is deposited on the storage layer **822** prior to the deposition of the storage layer **824**. As illustrated in FIG. 6B, the electrode **810**, the storage layer **822**, the blocking layer **826**, the storage layer **824**, and the electrode **830** are stacked in sequential order. That is, the composite storage layer **820** is sandwiched between the electrode **810** and the electrode **830**. [0059] In some embodiments, the electrode **810**, the storage layer **822**, the storage layer **824**, and the electrode **830** in FIG. 6B are respectively similar to the electrode **810**, the storage layer **822**, the storage layer **824**, and the electrode **830** in FIG. 6A, so the detailed descriptions thereof are omitted herein.

[0060] In some embodiments, a dielectric constant of the blocking layer **826** is greater than the dielectric constant of the storage layer **822** and the dielectric constant of the storage layer **824**. For example, the dielectric constant of the blocking layer **826** is greater than 10. In some embodiments, a crystallinity of the blocking layer **826** is also greater than a crystallinity of the storage layer **822** and a crystallinity of the storage layer **824**. In order to satisfy the foregoing constraint, a material of the blocking layer **826** is different from the material of the storage layer **822** and the material of the storage layer **824**. For example, the material of the blocking layer **826** includes AlO, Y.sub.2O.sub.3, HfO having a dopant, ZrO having a dopant, or a combination thereof. The dopant in the blocking layer **826** includes La, Al, Si, or a combination. In some embodiments, the dopant in the blocking layer **826** is the same as the dopant in the storage layer **822** and the dopant in the storage layer **824**. Nevertheless, a doping concentration of the dopant in the blocking layer **826** is greater than the doping concentration of the dopant in the storage layer **822** and the doping concentration of the dopant in the storage layer **824**. However, the disclosure is not limited thereto. In some alternative embodiments, the dopant in the blocking layer **826** is different from the dopant in the storage layer **822** and the dopant in the storage layer **824**. It should be noted that the material of the blocking layer **826** is not particular limited, as long as the dielectric constant and the crystallinity of the blocking layer **826** is greater than the dielectric constant and the crystallinity of the storage layer **822** and the dielectric constant and the crystallinity of the storage layer **824**.

[0061] In some embodiments, since the blocking layer **826** is sandwiched between the storage layer **822** and the storage layer **824**, the blocking layer **826** is able to serve as a crystalline interruption layer for the storage layers **822** and **824**. That is, the blocking layer **826** may effectively suppress the storage layers **822** and **824** from crystallization, thereby increasing the overall capacitance of the capacitor **800**.

[0062] Referring back to FIG. 5E, after the capacitor **800** is formed, a sub-dielectric layer **530**, an etch stop layer **900**, and a sub-dielectric layer **540** are sequentially formed on the capacitor **800** and the sub-dielectric layer **520**. In some embodiments, the sub-dielectric layer **530** and the sub-dielectric layer **540** are part of one of the dielectric layers **36** of the interconnect structure **30** in FIG. 1. In some embodiments, materials and a formation methods of the sub-dielectric layer **530** and the sub-dielectric layer **540** are similar to that of the sub-dielectric layer **510**, so the detailed descriptions thereof are omitted herein. As illustrated in FIG. 5E, the sub-dielectric layer **530** is in physical contact with the capacitor **800**. For example, the sub-dielectric layer **530** is in physical contact with the electrode **810**, the composite storage layer **820**, and the electrode **830**.

[0063] In some embodiments, the etch stop layer **900** is sandwiched between the sub-dielectric layer **530** and the sub-dielectric layer **540**. In some embodiments, a material of the etch stop layer **900** includes silicon carbide, silicon nitride, or the like. In some embodiments, the etch stop layer **900** is formed by any suitable method, such as CVD, spin-on coating, or the like.

[0064] Referring to FIG. 5F, a conductive feature **1000** is formed in the sub-dielectric layer **530**, the etch stop layer **900**, and the sub-dielectric layer **540** to obtain the capacitor structure **C2**. In some embodiments, the conductive feature **1000** includes a conductive via **1002** and a conductive line **1004** disposed on the conductive via **1002**. As illustrated in FIG. 5F, the conductive via **1002** is embedded in the sub-dielectric layer **530**. Meanwhile, the conductive line **1004** is embedded in the etch stop layer **900** and the sub-dielectric layer **540**. In some embodiments, the conductive feature **1000** includes a seed layer and a metal layer on the seed layer. The seed layer may be a conformal layer lining sidewalls and a bottom surface of the metal layer. In some embodiments, the seed layer may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. For example, the seed layer is a titanium/copper composited layer. The metal layer may include a metal material, such as Al, Cu, W, Co, Pd, Pt, Ni, other low resistivity metal constituent, an alloy thereof, or a combination thereof, and are formed by an electroplating process. In some embodiments, the conductive feature **1000** is electrically connected to the conductive patterns **34** and the conductive vias **32** of the interconnect structure **30** in FIG. 1.

[0065] As illustrated in FIG. 5F, the conductive via **1002** of the conductive feature **1000** is in physical contact with the capacitor **800**. For example, the conductive via **1002** of the conductive feature **1000** is in physical contact with the electrode **830**. As such, the conductive feature **1000** is electrically connected to the capacitor **800**. Moreover, the capacitor **800** is in physical contact with the conductive pattern **600**. For example, the electrode **810** is in physical contact with the conductive pattern **600**. As such, the conductive pattern **600** is electrically connected to the capacitor **800**. That is, the capacitor **800** is connected to the conductive patterns **34** and the conductive vias **32** of the interconnect structure **30** in FIG. 1 through the conductive feature **1000** and/or the conductive pattern **600**.

[0066] In some embodiments, since the interconnect structure **30** is being considered as formed during back-end-of-line (BEOL) process, the capacitor structure **C2** is also being considered as formed during BEOL processes.

[0067] FIG. 7 is a schematic cross-sectional view of an integrated circuit **IC3** in accordance with some alternative embodiments of the disclosure.

[0068] Referring to FIG. 7, the integrated circuit **IC3** in FIG. 7 is similar to the integrated circuit **IC1** in FIG. 1, so the detailed description thereof is omitted herein. The difference between the integrated circuit **IC3** in FIG. 7 and the integrated circuit **IC1** in FIG. 1 lies in that the integrated circuit **IC3** in FIG. 7 further includes a capacitor structure **C2**. As illustrated in FIG. 7, the capacitor structure **C2** is embedded in the interconnect structure **30**. For example, the capacitor structure **C2** is embedded in one of the dielectric layers **36**. In some embodiments, the capacitor structure **C2** is electrically connected to the underlying and overlying conductive vias **32**/conductive patterns **34**. As illustrated in FIG. 7, the capacitor structure **C2** is disposed on top of the capacitor structure **C1**. In some embodiments, the capacitor structure **C1** and the capacitor structure **C2** are embedded in different dielectric layers **36**. In some embodiments, the capacitor structure **C2** in FIG. 7 may be similar to the capacitor structure **C2** in FIG. 4 and FIG. 5F, so the detailed description thereof is omitted herein. As mentioned above, the capacitor structure **C1** includes the capacitor **300** and the capacitor structure **C2** includes the capacitor **800**. As such, the capacitor **300** in FIG. 2F and the capacitor **800** in FIG. 5F coexist in the integrated circuit **IC3**. For example, the capacitor **800** is disposed over the capacitor **300**.

[0069] In some embodiments, since the interconnect structure **30** is being considered as formed during back-end-of-line (BEOL) process, the capacitor structure **C1** and the capacitor structure **C2** are also being considered as formed during BEOL processes.

[0070] In accordance with some embodiments of the disclosure, a capacitor includes a first electrode, a second electrode over the first electrode, and a composite storage layer sandwiched between the first electrode and the second electrode. The composite storage layer includes a first storage layer and a second storage layer. The first storage layer is in physical contact with the first

electrode. The second storage layer is in physical contact with the second electrode. A conduction band offset of the first storage layer is greater than a conduction band offset of the second storage layer, and a dielectric constant of the second storage layer is greater than a dielectric constant of the first storage layer.

[0071] In accordance with some embodiments of the disclosure, an integrated circuit includes a substrate, a transistor, and an interconnect structure. The transistor is over the substrate. The interconnect structure is disposed on the substrate and includes dielectric layers and a first capacitor. The first capacitor is embedded in the dielectric layers. The first capacitor includes a first electrode, a first storage layer, a second electrode, and a second storage layer. The first storage layer is on the first electrode. The second electrode is over the first storage layer. The second storage layer is between the first storage layer and the second electrode. A material of the first storage layer is different from a material of the second storage layer.

[0072] In accordance with some embodiments of the disclosure, a manufacturing method of an integrated circuit includes at least the following steps. A substrate is provided. A transistor is formed over the substrate. An interconnect structure is formed over the substrate. The interconnect structure is formed by at least the following steps. First, dielectric layers are formed. Thereafter, a capacitor is formed in the dielectric layers. The capacitor is formed by at least the following steps. A first electrode is deposited. A composite storage layer is formed on the first electrode. A second electrode is deposited on the composite storage layer. The composite storage layer is formed by at least the following steps. A first storage layer is deposited on the first electrode. A second storage layer is deposited over the first storage layer. A material of the first storage layer is different from a material of the second storage layer.

[0073] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A capacitor, comprising: a first electrode; a second electrode over the first electrode; and a composite storage layer sandwiched between the first electrode and the second electrode, comprising: a first storage layer in physical contact with the first electrode; and a second storage layer in physical contact with the second electrode, wherein a conduction band offset of the first storage layer is greater than a conduction band offset of the second storage layer, and a dielectric constant of the second storage layer is greater than a dielectric constant of the first storage layer.
2. The capacitor of claim 1, wherein a first voltage is applied to the first electrode, a second voltage is applied to the second electrode, and the second voltage is greater than the first voltage.
3. The capacitor of claim 2, wherein the first voltage is a negative voltage and the second voltage is a positive voltage.
4. The capacitor of claim 1, wherein a material of the first storage layer comprises LaO, AlO, SiO, HfO having a dopant, ZrO having a dopant, or a combination thereof.
5. The capacitor of claim 4, wherein a material of the second storage layer comprises ZrO, HfO, HfZrO, HfO having a dopant, ZrO having a dopant, or a combination thereof.
6. The capacitor of claim 5, wherein the dopant in the first storage layer and the dopant in the second storage layer respectively comprise La, Al, Si, or a combination thereof.
7. The capacitor of claim 5, wherein a doping concentration of the dopant in the first storage layer

is greater than a doping concentration of the dopant in the second storage layer.

8. The capacitor of claim 1, wherein the composite storage layer further comprises a blocking layer sandwiched between the first storage layer and the second storage layer.

9. The capacitor of claim 8, wherein a material of the blocking layer comprise AlO, Y.sub.2O.sub.3, HfO having a dopant, ZrO having a dopant, or a combination thereof.

10. An integrated circuit, comprising: a substrate; a transistor over the substrate; and an interconnect structure disposed on the substrate, comprising: dielectric layers; and a first capacitor embedded in the dielectric layers, comprising: a first electrode; a first storage layer on the first electrode; a second electrode over the first storage layer; and a second storage layer between the first storage layer and the second electrode, wherein a material of the first storage layer is different from a material of the second storage layer.

11. The integrated circuit of claim 10, wherein a conduction band offset of the first storage layer is greater than a conduction band offset of the second storage layer, and a dielectric constant of the second storage layer is greater than a dielectric constant of the first storage layer.

12. The integrated circuit of claim 10, wherein a negative voltage is applied to the first electrode and a positive voltage is applied to the second electrode.

13. The integrated circuit of claim 10, wherein a material of the first storage layer comprises LaO, AlO, SiO, HfO having a dopant, ZrO having a dopant, or a combination thereof, and a material of the second storage layer comprises ZrO, HfO, HfZrO, HfO having a dopant, ZrO having a dopant, or a combination thereof.

14. The integrated circuit of claim 13, wherein the dopant in the first storage layer and the dopant in the second storage layer respectively comprise La, Al, Si, or a combination thereof.

15. The integrated circuit of claim 10, wherein the first capacitor further comprises a blocking layer sandwiched between the first storage layer and the second storage layer.

16. The integrated circuit of claim 15, wherein a material of the blocking layer comprise AlO, Y.sub.2O.sub.3, HfO having a dopant, ZrO having a dopant, or a combination thereof.

17. The integrated circuit of claim 10, wherein the interconnect structure further comprises a second capacitor over the first capacitor and embedded in the dielectric layers, and the second capacitor comprises: a third electrode; a third storage layer on the third electrode; a fourth electrode over the third storage layer; and a fourth storage layer between the third storage layer and the fourth electrode, wherein a conduction band offset of the third storage layer is greater than a conduction band offset of the fourth storage layer, and a dielectric constant of the fourth storage layer is greater than a dielectric constant of the third storage layer.

18. A manufacturing method of an integrated circuit, comprising: providing a substrate; forming a transistor over the substrate; and forming an interconnect structure over the substrate, comprising: forming dielectric layers; and forming a capacitor in the dielectric layers, comprising: depositing a first electrode; forming a composite storage layer on the first electrode, comprising: depositing a first storage layer on the first electrode; and depositing a second storage layer over the first storage layer, wherein a material of the first storage layer is different from a material of the second storage layer; and depositing a second electrode on the composite storage layer.

19. The method of claim 18, wherein a conduction band offset of the first storage layer is greater than a conduction band offset of the second storage layer, and a dielectric constant of the second storage layer is greater than a dielectric constant of the first storage layer.

20. The method of claim 18, wherein forming the composite storage layer further comprises: depositing a blocking layer on the first storage layer prior to the deposition of the second storage layer.
